

CONTACT

INFORMATION

Ph.D. in Mechanical Engineering
Seoul National University, Seoul, Korea

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RESEARCH
INTERESTS

CAE automation and AI-assisted simulation, multi-physics modeling of heat/moisture diffusion and deformation, low Re fluid dynamics

EDUCATION

Seoul National University, Seoul, Korea

Integrated M.S. and Ph.D. , Mechanical Engineering,

03/2020 to 02/2026

- Advisor: Professor Ho-Young Kim
- Area of study:

Heat- and moisture-transport-driven material deformation, fluid transport in HAR nano patterns and porous media, hydrogel synthesis and adhesion

Seoul National University, Seoul, Korea

B.S., Mechanical Engineering (including military service)

03/2014 to 02/2020

SKILLS

- CAE: ABAQUS, COMSOL, Hypermesh, motion-view
- AFM, SAXS, E-SEM, E-TEM, Electro-spinning
- Polymer synthesis: hydrogel crosslinking and interfacial bonding with other materials

PAPERS

1. C. Kim^{*}, J. Won^{*}, D. Kim^{*}, S. Jung[†], H.-Y. Kim[†], and Y. Hyun[†]. *Viola seed pod architecture shapes sequential, force-augmented pinching* (2026). *Science* (IF: 45.8).
2. H. Jeong^{*}, C. Kim^{*}, A. Lee[†], and H.-Y. Kim[†] Sequential multi-modal morphing of single-input pneu-nets (2023). *Soft Robotics* (IF: 6.4, Q1 in Robotics based on JCR).
3. S. Jung^{*}, C. Kim^{*}, T. Kim, S. Park, and H.-Y. Kim[†] Mechanics of water-powered soft actuation (2025). *Chemical Reviews* (IF: 51.4, Invited) – 2nd revision.

* : Equal contribution, †: Corresponding author

CONFERENCE

PRESENTATIONS

1. H. Jeong, C. Kim, J. Ha, S. Lee, A. Lee and H.-Y. Kim, , Dual curvature control using pneumatic actuators, In: American Physical Society March Meeting 2021 (2021)
2. C. Kim, H. Jeong, J. Ha, S. Lee, A. Lee and H.-Y. Kim, Dual morphing pneu-net by stiffness control, In: Korean Society of Mechanical Engineers autumn conference 2021 (2021)
3. C. Kim, H. Jeong, J. Ha, A. Lee and H.-Y. Kim, Multi-morph pneu-net systems with a single input, In: American Physical Society March Meeting 2022 (2022)
4. H. Jeong, C. Kim, J. Ha, P. Woo, A. Lee and H.-Y. Kim, Dual curvature control using pneumatic actuators, In: The Second International Conference on Nature-Inspired Surface Engineering 2022 (2022) - Best presentation prize
5. C. Kim, S. Jung and H.-Y. Kim, Dual morphing hydrogel with a fibrous membrane, In: Korean Society of Mechanical Engineers autumn conference 2023 (2023)
6. C. Kim and H.-Y. Kim, Sequential multimodal morphing of single-input pneu-nets, In: Soft Robotics invited talk 2024 (2024)

7. C. Kim, J. Won, S. Jung, Y. Hyun and H.-Y. Kim**, Humidity-responsive poroelastic zippering of Viola seed pods, In: IUTAM symposium on capillarity and elastocapillarity in Biology 2024 (2024), **: invited speaker
8. C. Kim, J. Won, S. Jung, Y. Hyun and H.-Y. Kim, Poroelastic zippering actuation of hygroresponsive Viola seed pods, In: American Physical Society Division of Fluid 2024 (2024)
9. H. Kim, C. Kim, H. Jeon, C. Park, S.Jung and H.-Y. Kim, Mechano-Osmosis Induced Turgor in Hydrogel-Encapsulated Liquid Pouch, In: NCFE 14 (2026)
10. M. Han C. Kim, J. Lee, JM. Lee, H.Jung, S. Je, Yamaoka, Y., H.-Y. Kim and Y. Lee De novo cuticle formation acts as a multiscale mechanical system controlling growth and robustness after floral organ abscission, In: American Society of Plant Biologists (ASPB) (2026)

PATENTS

1. C. Kim, H. Jeong, J. Lee and H.-Y. Kim, Pneumatic valve device using expansion of elastic material protruding parts. 10-2397568 KOREA (2022)
2. C. Kim, H. Jeong and H.-Y. Kim, Pneu-net device capable of sequential dual deformation. 10-2478112 KOREA (2022) - granted
3. C. Kim, H. Jeong and H.-Y. Kim, Pneu-net device capable of sequential torsional and bending deformation. 10-2478131 KOREA (2022) - granted
4. C. Kim, H. Jeong, S. O, J. Nam and H.-Y. Kim, Massage device and controlling method thereof. 1020200154180 KOREA (2020)
5. C. Kim, H. Jeong, S. O, J. Nam, J. In and H.-Y. Kim, Massage device having twist function and control method thereof. 1020210149522 KOREA (2021)
6. C. Kim, H. Jeong, G. Lee, H. Jeong, W. Park and H.-Y. Kim, TUBE CONTAINER. 1020220039765 KOREA (2022)
7. C. Kim, H. Jeon, J. kim and H.-Y. Kim, Plumbing apparatus containing hydrogel part. 1020230186336 KOREA (2023)
8. C. Kim, H. Jeon and H.-Y. Kim, Method for Manufacturing grooved surface using mesh structure. 1020230186335 KOREA (2023)
9. C. Kim, J. Won, D.Kim, S. Jung, H.-Y. Kim and Hyun Y, Biomedical devices inspired by plant morphing mechanisms. 1020260015498 KOREA (2026)

HONORS AND
AWARDS

Open innovation challenge encouragement prize, KSME-SEMES	12/2020
Future home tech challenge encouragement prize, KSME-LG	10/2023
Future home tech challenge silver award, KSME-LG	11/2024

RESEARCH
PROJECTS

Research center for metamorphing mechanical systems, Awarding body: National Research Foundation of Korea,	03/2020 - 02/2026
Developing a pneu-net system for personalized massager, Awarding body: Amore Pacific. Inc.,	06/2021 - 09/2021
Developing a novel origami structure for packaging, Awarding body: Amore Pacific. Inc.,	06/2022 - 10/2022

Developing automatic humidity control system using hygroscopic materials,

Awarding body: LG electronics

07/2023 - 02/2026

Study on leaning and bending models in composite nano structures,

Awarding body: SAMSUNG electronics

10/2023 - 09/2025

ACADEMIC MEMBERSHIPS	• American Physical Society (APS), Member,	2021-2025
	• Korean Society of Mechanical Engineers (KSME), Member,	2021-2025

REFERENCES	Ho-Young Kim, PhD	
	• Professor, Department of Mechanical Engineering, Seoul National University, Seoul, Korea	
	• Relationship: PhD advisor	
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